MINI PROJECT 1

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Domain: Data science

Iris dataset

The Iris dataset is contains samples of iris flowers across three variety—Setosa, Versicolor, and Virginica—with four measured features: sepal length, sepal width, petal length, and petal width

Use of Linear regression

We upload the dataset in jupyter notebook

```
[1]: import pandas as pd
```

```
[2]: iris=pd.read_csv('iris.csv')
```

```
[4]: iris.head()
```

[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

This shows the top 5 data in the dataset this basically shows that the dataset is uploaded and read

```
[8]: iris[iris['sepal.width']>4]
```

[8]:

	sepal.length	sepal.width	petal.length	petal.width	variety
15	5.7	4.4	1.5	0.4	Setosa
32	5.2	4.1	1.5	0.1	Setosa
33	5.5	4.2	1.4	0.2	Setosa

This shows the data where sepal width >4 the variety is setosa

```
[12]: iris[iris['petal.width']>1]
```

[12]:

	sepal.length	sepal.width	petal.length	petal.width	variety
50	7.0	3.2	4.7	1.4	Versicolor
51	6.4	3.2	4.5	1.5	Versicolor
52	6.9	3.1	4.9	1.5	Versicolor
53	5.5	2.3	4.0	1.3	Versicolor
54	6.5	2.8	4.6	1.5	Versicolor
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

93 rows × 5 columns

Here data with petal width >1 is shown which implies tht versicolor and virginica have petal width>1

```
[13]: iris[iris['petal.width']>2]
```

[13]:

	sepal.length	sepal.width	petal.length	petal.width	variety
100	6.3	3.3	6.0	2.5	Virginica
102	7.1	3.0	5.9	2.1	Virginica
104	6.5	3.0	5.8	2.2	Virginica
105	7.6	3.0	6.6	2.1	Virginica
109	7.2	3.6	6.1	2.5	Virginica
112	6.8	3.0	5.5	2.1	Virginica
114	5.8	2.8	5.1	2.4	Virginica
115	6.4	3.2	5.3	2.3	Virginica
117	7.7	3.8	6.7	2.2	Virginica
118	7.7	2.6	6.9	2.3	Virginica
120	6.9	3.2	5.7	2.3	Virginica
124	6.7	3.3	5.7	2.1	Virginica
128	6.4	2.8	5.6	2.1	Virginica
132	6.4	2.8	5.6	2.2	Virginica
135	7.7	3.0	6.1	2.3	Virginica
136	6.3	3.4	5.6	2.4	Virginica
139	6.9	3.1	5.4	2.1	Virginica
140	6.7	3.1	5.6	2.4	Virginica
141	6.9	3.1	5.1	2.3	Virginica
143	6.8	3.2	5.9	2.3	Virginica
144	6.7	3.3	5.7	2.5	Virginica
145	6.7	3.0	5.2	2.3	Virginica
148	6.2	3.4	5.4	2.3	Virginica

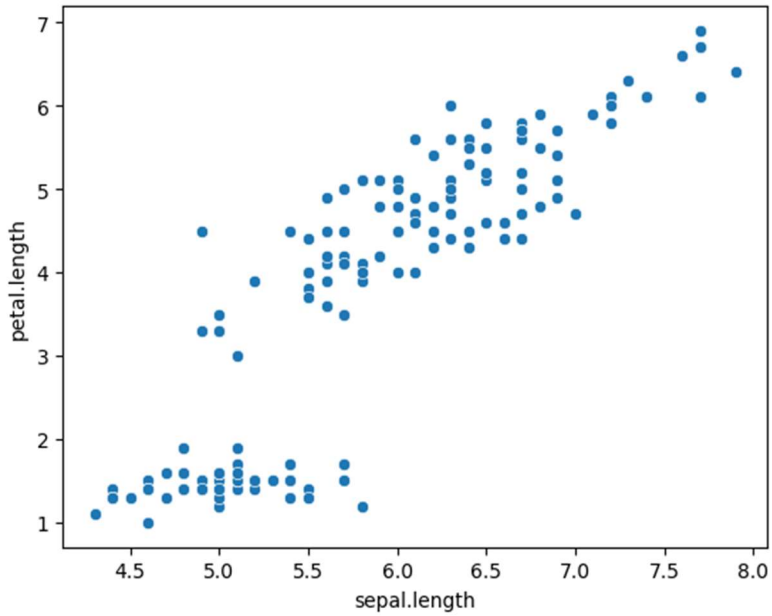
```
[14]: from matplotlib import pyplot as plt
```

The variety with petal width>2 is virginica

```
[14]: from matplotlib import pyplot as plt
```

```
[15]: import seaborn as sns
```

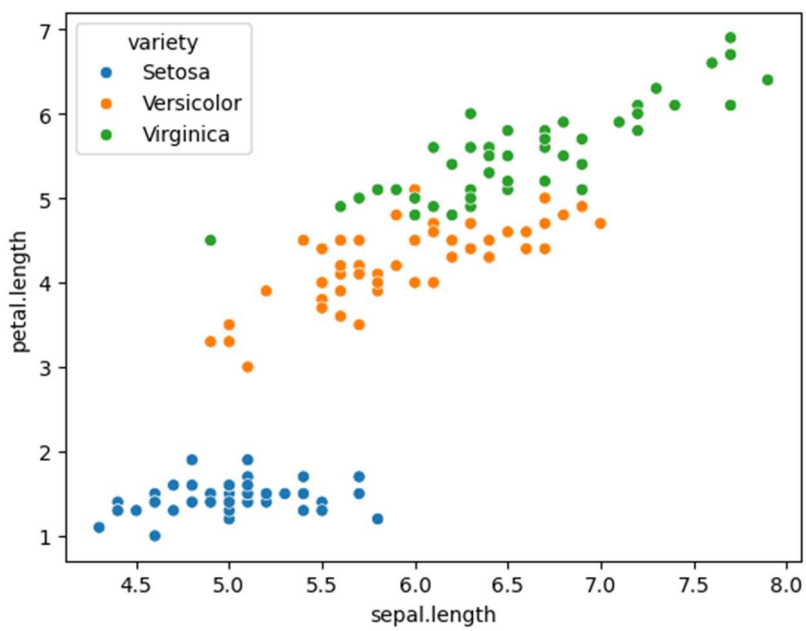
```
[16]: sns.scatterplot(x='sepal.length',y='petal.length',data=iris)
      plt.show()
```



Here we see the scatter plot of sepal length and petal length

Next we see the plot of different types of variety in iris flower

```
[20]: sns.scatterplot(x='sepal.length',y='petal.length',data=iris,hue='variety')
      plt.show()
```



```
[22]: y=iris[['sepal.length']]

[23]: x=iris[['sepal.width']]

[24]: from sklearn.model_selection import train_test_split;

[25]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)

[26]: x_train.head()
```

[26]:

	sepal.width
34	3.1
1	3.0
20	3.4
103	2.9
124	3.3

```
[27]: x_test.head()
```

[27]:

	sepal.width
82	2.7

splitting the Iris dataset into **training data (70%) and testing data (30%)**, where sepal width is the input feature and sepal length is the target output using the train_test_split()

here

```
[26]: x_train.head()
```

[26]:

	sepal.width
34	3.1
1	3.0
20	3.4
103	2.9
124	3.3

\

Xtrain is training data 1st 5 data is shown

[27]:
```
x_test.head()
```

[27]:

	sepal.width
82	2.7
25	3.0
144	3.3
13	3.0
135	3.0

[28]:
```
y_train.head()
```

[28]:

	sepal.length
34	4.9
1	4.9
20	5.4
103	6.3
124	6.7

---

[29]:
```
y_test.head()
```

[29]:

	sepal.length
82	5.8
25	5.0
144	6.7
13	4.3
135	7.7

[30]:
```
#categorize the data was done till here as train and test data
```

[31]:
```
from sklearn.linear_model import LinearRegression
```

[32]:
```
lr=LinearRegression()
```

[33]:
```
lr.fit(x_train,y_train)
```

[33]:
```
▾ LinearRegression
LinearRegression()
```

[34]:
```
y_pred=lr.predict(x_test)
```

Here we use linear regression and fit it with xtrain and ytrain and based on this we predict on xtest

```
[34]: y_pred=lr.predict(x_test)
```

```
[35]: y_test.head()
      #actual value
```

[35]:

	sepal.length
82	5.8
25	5.0
144	6.7
13	4.3
135	7.7

```
[36]: y_pred[0:5]
      #predicted value
```

```
[36]: array([[5.97441271],
             [5.86965818],
             [5.76490365],
             [5.86965818],
             [5.86965818]])
```

```
[46]: #we find out the error in prediction
      from sklearn.metrics import mean_squared_error
      mean_squared_error(y_test,y_pred)
```

```
[46]: 0.805276941542795
```

Here the accuracy is shown in the range of 57 to 59 % this is on two independent variable and we see the error in prediction is 0.805276941542795

So we use model 2 using more than 2 independent variable

Here we use 3 independent variables

model 2

```
[38]: #using 3 independent values

[45]: y=iris[['sepal.length']]

[44]: x=iris[['sepal.width','petal.length','petal.width']]

[47]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)

[49]: lr2=LinearRegression()

[50]: lr2.fit(x_train,y_train)

[50]: ▾ LinearRegression
      LinearRegression()

[53]: y_pred=lr2.predict(x_test)

[55]: y_test.head()
```

[55]:

	sepal.length
55	5.7
91	6.1
75	6.6
20	5.4
149	5.9

Here we use sepal width,petal length,petal width and use linear regression

here this proves that model 2 is more accurate compared to model 1

mean squared error must be small performance improves so error is less

model 2 gives better prediction than model 1

[53]:
```
y_pred=lr2.predict(x_test)
```

[55]:
```
y_test.head()
```

[55]:

	sepal.length
55	5.7
91	6.1
75	6.6
20	5.4
149	5.9

[56]:
```
y_pred[0:5]
```

[56]:
```
array([[6.14605368],
       [6.30780729],
       [6.15764553],
       [5.16115413],
       [6.43825543]])
```

[54]:
```
mean_squared_error(y_test,y_pred)
```

[54]:
```
0.09678672065880599
```